

Indian Statistical Institute

BSDS - Second Year

Exercise Set 1.

1. Let P, P_1 and P_2 be pgfs.
 - a) Show that $\lambda P_1 + (1 - \lambda)P_2$ is a pgf for any $0 \leq \lambda \leq 1$. Show that $P_1 P_2$ is a pgf.
 - b) Is $\exp(P(s) - 1)$ a pgf? Fix $0 < \lambda < 1$. Is $P(\lambda s)/P(\lambda)$ necessarily a pgf?
2. Find the pgf for the following random variables and state their radius of convergence:
 - a) X where $\mathbb{P}(X = n) = \frac{c}{n!}$ for $n \geq 0$. Find the constant c first.
 - b) Y where $\mathbb{P}(Y = n) = \frac{c}{(n+1)(n+2)}$ for $n \geq 0$. Find the constant c first.
 - c) Z where $\mathbb{P}(Z = n) = \frac{1}{2^{n+1}}$ for $n \geq 0$.
3. Find the distribution of the random variables in each of the cases and find $\mathbb{P}(X \geq n)$ for $n \geq 1$.
 - a) $P(s) = \frac{1}{3-2s}$;
 - b) $P(s) = \frac{1}{4-3s}$.
 - c) Two dice are rolled. Let X be the sum of the outcomes. Find the pgf of X . Use this pgf to calculate the probability that the sum is divisible by 3.
 - d) If the pgf of a random variable X is $P(s) = \frac{1}{2}e^s + \frac{1}{4}s^2 + \frac{1}{4}$, find $\mathbb{P}(X \geq 2)$.
4.
 - a) Use generating functions to prove the combinatorial identity: $\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$.
 - b) Find the sequence whose generating function is $\frac{1}{(1-s)^4}$ by expressing it as a product of known generating functions and using convolution.
 - c) Let a_n be the number of ways to tile a $2 \times n$ board with 1×2 dominoes for $n \geq 1$. Find a recurrence relation for a_n for $n \geq 3$. Set a_1, a_2 appropriately and, find its generating function.

5. a) Let p_k be the probability that three consecutive Heads (HHH) is obtained for the first time after the k -th toss of a coin. Find the generating function of $\{p_k : k \geq 0\}$. Compute the expected time for this to happen.
- b) Let p_k be the probability of the pattern of two consecutive Heads (HH) or the pattern of Head followed by Tail (HT) is obtained for the first time after the k -th toss of a coin. Find the generating function of $\{p_k : k \geq 0\}$. Compute the expected time for this to happen.
- c) Let p_k be the probability that the total obtained in k throws of a dice (not necessarily fair) is divisible by 3. Find the generating function of $\{p_k : k \geq 0\}$. Find $\lim_{k \rightarrow \infty} p_k$.
6. a) Claims to an insurance company arrive as a Poisson process with rate λ . The size of each claim is an **iid** random variable Y with pgf $P_Y(s)$. Let S_n be the total claim amount up to time n . Find the pgf of S_n .
- b) A hen lays N eggs, where $N \sim \text{Poisson}(\lambda)$. Each egg hatches with probability p independently. Let K be the number of chicks. Find the pgf of K and identify its distribution.
- c) In the previous problem, find the expected number of chicks, $\mathbb{E}[K]$, using the random sum formula and verify by direct computation.

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Exercise Set 2.

1. Consider a branching process starting with a single individual, $Z_0 = 1$.
 - a) If the offspring distribution is given by $\mathbb{P}(\xi = 0) = 1/4$, $\mathbb{P}(\xi = 1) = 1/2$, $\mathbb{P}(\xi = 2) = 1/4$, compute $\mathbb{E}(Z_3)$ and $\text{Var}(Z_3)$. Also, find the probability that $\mathbb{P}(Z_3 = 0)$.
 - b) If the offspring distribution is given by $\mathbb{P}(\xi = 0) = 1/6$, $\mathbb{P}(\xi = 1) = 1/2$, $\mathbb{P}(\xi = 2) = \mathbb{P}(\xi = 3) = 1/6$, compute $\mathbb{E}(Z_2 Z_3)$ and $\text{Var}(Z_2 + Z_3)$. Also, find the probability $\mathbb{P}(Z_2 > 0, Z_3 = 0)$.

2. Suppose that $\{Z_n : n \geq 0\}$ is a branching process with $Z_0 = 1$. Fix any positive integer $r \geq 1$ and define $\{Y_n : n \geq 0\}$ as follows:

$$Y_n = Z_{nr} \text{ for } n \geq 0.$$

Check that $\{Y_n : n \geq 0\}$ is also a branching process. Find its probability generating function of the offspring distribution.

3. Consider a branching process with generating function of the offspring pgf given by $G(s) = q/(1 - ps)$ where $0 < p < 1$, $q = 1 - p$.

- a) For $p = 1/2$, show that

$$G_n(s) = \frac{(n - (n - 1)s)}{(n + 1 - ns)}$$

for $n \geq 1$. Show that $G_n(0) \rightarrow 1$ as $n \rightarrow \infty$.

- b) For $p \neq 1/2$, show that

$$G_n(s) = q \frac{p^n - q^n - (p^{n-1} - q^{n-1})ps}{p^{n+1} - q^{n+1} - (p^n - q^n)ps}.$$

Further conclude that

$$\lim_{n \rightarrow \infty} G_n(0) = \begin{cases} 1 & \text{if } p < q \\ \frac{q}{p} & \text{if } p > q. \end{cases}$$

4. Let $\{Z_n\}$ be a branching process with $Z_0 = 1$ and offspring mean μ .

a) Assume that $G'''(1) < \infty$. Show that $\mathbb{E}(Z_n Z_m) = \mu^{n-m} \mathbb{E}(Z_m^2)$ for $m \leq n$. Hence find the covariance and the correlation coefficient between Z_m and Z_n (in terms of μ) for $m \leq n$.

b) Find $\mathbb{E}(Z_n | Z_m = k)$ for $m < n$ and $k \geq 0$.

5. Find the extinction probability ρ for a branching process with offspring distribution:

a) i) $\mathbb{P}(\xi = 0) = 2/3, \mathbb{P}(\xi = 3) = 1/3$.

ii) $\mathbb{P}(\xi = 0) = 1/2, \mathbb{P}(\xi = 1) = \mathbb{P}(\xi = 3) = 1/4$.

b) Consider a branching process where each individual has a random number of offspring with a $\text{Poisson}(\lambda)$ distribution. Determine the critical value λ_c below which the branching process becomes extinct with probability 1. Find the extinction probability ρ for $\lambda > \lambda_c$.

c) Suppose that the probability generating function of the offspring distribution is given by

$$G(s) = p_0 + p_1 s + p_2 s^2$$

with $p_2 > 0, p_k = 0$ for all $k \geq 3$ and $2p_2 + p_1 > 1$. Show that the extinction probability is given by

$$\rho = \frac{p_0}{p_2}.$$

d) A branching process starts with $Z_0 = N$ individuals, each evolving independently with the same offspring distribution. Show that the probability of ultimate extinction is ρ^N , where ρ is the extinction probability starting with one individual.

6*. We set the *total progeny* of the branching process as

$$S = \sum_{n=0}^T Z_n \tag{1}$$

where $Z_0 = 1$ and T is the extinction time. We assume that $T < \infty$ with probability 1, i.e., the family becomes extinct with probability 1.

a) Show that the generating function of S is given by the solution of the following equation

$$H(s) = sG(H(s))$$

for $0 < s \leq 1$, where the probability generating function of the offspring distribution is given by $G(s)$. Show that the above equation admits a unique solution when $0 \leq s \leq 1$.

- b) Show that the total progeny of the branching process has the generating function given by

$$H(s) = \frac{1}{2ps} (1 - \sqrt{1 - 4pqs^2})$$

when the probability generating function of the offspring distribution is given by $G(s) = q + ps^2$ where $p \leq 1/2$.

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Exercise Set 3.

1. Define for $a, b > 0$, $N_n^{a,b}$ as the number of paths $\pi = \{0 = s_0, s_1, s_2, \dots, s_n = a\}$ with $|s_i - s_{i-1}| = 1$, (starting from $(0, 0)$ to (n, a)) such that $s_i < a$ and $s_i > -b$ for all $i = 1, 2, \dots, n-1$. Note a, b can be ∞ also.

a) Show that

$$N_n^{a,b} = N_n^{a,\infty} - N_n^{a+b,\infty}.$$

b) Find $N_n^{a,\infty}$ and $N_n^{a+b,\infty}$.

2. For $n \geq 1$, show that

a)

$$u_{2n} = (-1)^n \binom{-\frac{1}{2}}{n}$$

and hence show that the generating function $U(s) = \sum_{n=0}^{\infty} u_n s^n$ is given by

$$U(s) = \frac{1}{\sqrt{1-s^2}}.$$

b)

$$f_{2n} = \frac{1}{2n-1} u_{2n} = (-1)^{n-1} \binom{\frac{1}{2}}{n}$$

and hence show that the generating function $F(s) = \sum_{n=0}^{\infty} f_n s^n$ is given by

$$F(s) = 1 - \sqrt{1-s^2}.$$

3. For $n \geq 1$, show that

a)

$$\mathbb{P}(S_1 \geq 0, S_2 \geq 0, \dots, S_{2n} \geq 0) = u_{2n}.$$

b)

$$\mathbb{P}(S_1 \geq -1, \dots, S_{2n-1} \geq -1) = 2u_{2n}.$$

c)

$$\mathbb{P}(S_1 \geq 0, \dots, S_{2n-1} \geq 0, S_{2n} = 0) = 2f_{2n+2}.$$

4.

a) Show that the expected number of returns to the origin in n steps is $\sum_{k=0}^n u_k \sim \frac{2}{\pi} \sqrt{n}$ as $n \rightarrow \infty$.

b) For $b \neq 0$, show that

$$\mathbb{P}(S_1 \neq 0, S_2 \neq 0, \dots, S_n \neq 0, S_n = b) = \frac{|b|}{n} \mathbb{P}(S_n = b).$$

Show that

$$\mathbb{P}(S_{2n} = 0) = \frac{1}{2n} \mathbb{E}(|S_{2n}|).$$

Show that, for $b \geq 0$,

$$\mathbb{P}(\max\{S_0, S_1, \dots, S_n\} = b) = \mathbb{P}(S_n = b) + \mathbb{P}(S_n = b + 1).$$

c) Let $M_n = \max\{0, S_1, \dots, S_n\}$. For any $k > 0$, show that

$$\mathbb{P}(S_{2n} = 0, M_{2n-1} = k) = \mathbb{P}(S_{2n} = 2k).$$

5. For a simple symmetric random walk, show that

$$\mathbb{P}(\text{Infinitely Many returns to Origin}) = 1$$

6*. We say the particle spends the time interval $(i, i + 1)$ above the x-axis if at least one of the time points i and $i + 1$ the value is larger than 0. Otherwise, we say that the particle stays below the x-axis. Define

$$L_{2n} = \sum_{i=1}^{2n} \mathbb{I}\{\text{the particle spends } (i-1, i) \text{ is above x-axis}\}.$$

For $n \geq 1$, show that

$$\mathbb{P}(L_{2n} = 2k) = u_{2k} u_{2n-2k}.$$

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Exercise Set 4.

1. Let $\{S_n : n \geq 0\}$ be the simple symmetric ($p = 1/2$) random walk in one dimension with $S_0 = 0$.

- a) Show that $\{|S_n| : n \geq 0\}$ a Markov chain on $\{0, 1, 2, \dots\}$. Find its transition matrix.
- b) Let $M_n = \max\{S_k : 0 \leq k \leq n\}$, and show that $Y_n = M_n - S_n$ defines a Markov chain. Find the transition probabilities of this chain.

What happens if the random walk is simple but asymmetric? What happens if the starting position is not the origin, i.e., $S_0 \neq 0$.

2. A die is rolled repeatedly. The results be denoted by $\{\xi_n : n \geq 1\}$. Which of the following are Markov chains? For those that are, find the transition matrix.

- a) The largest number $\{M_n : n \geq 0\}$ shown up to the $(n+1)$ -th roll, i.e., $M_n = \max\{\xi_1, \dots, \xi_{n+1}\}$.
- b) The number N_n of sixes in n rolls, i.e., $N_n = \sum_{k=1}^n \mathbb{I}(\epsilon_{k+1} = 6)$.
- c) The number of throws since the last six, i.e., $L_n = n - \max\{k \leq n : \epsilon_{k+1} = 6\}$, with the convention that if the set $\{k \leq n : \epsilon_{k+1} = 6\}$ is empty, the maximum is taken to be 0.
- d) The number of throws until the next six, $R_n = \min\{k > n : \epsilon_{k+1} = 6\} - n$.

3. Suppose that $\{\xi_n : n \geq 1\}$ denote the scores obtained at the n -th throw of a fair dice. Set $S_n = (\epsilon_{n+1} + \epsilon_{n+2}) \bmod 6$ for $n \geq 0$. Show that $\{S_n : n \geq 1\}$ is a Markov Chain. Find the transition probability matrix for this chain.

4. Let $\{X_n : n \geq 0\}$ be a Markov chain on the state space $\mathcal{S}$ and transition matrix $\mathbf{P}$. Set, for $k \geq 1$ and $n \geq 0$,

$$Y_n := (X_n, X_{n+1}, \dots, X_{n+k}). \quad (1)$$

Show that the process $\{Y_n : n \geq 0\}$, taking values $\mathcal{S}^{k+1}$, is a Markov chain. What is its transition matrix? The Markov chain $\{Y_n : n \geq 0\}$ is often referred as the snake chain of length $k+1$ associated with $\{X_n : n \geq 0\}$.

5. Consider the chain $\{X_n : n \geq 0\}$, where X_n 's represents the result of independent tosses of a coin with probability of success being p . For $k = 2$, we sample space consists of all possible patterns of length 3, i.e., $\{(H,H,H), (H,H,T), (H,T,H), (H,T,T), (T,H,H), (T,H,T), (T,T,H), (T,T,T)\}$ (we will drop the separating comma in future). Find the transition matrix this snake chain.

6. Let $\{X_n^{(j)} : n \geq 0\}$ be independent sequences of Markov chains for $1 \leq j \leq k$ on the state space $\mathcal{S}_j$ and transition matrix $\mathbf{P}_j$. Set, for $n \geq 0$,

$$Y_n := (X_n^{(1)}, X_n^{(2)}, \dots, X_n^{(k)}). \quad (2)$$

Show that the process $\{Y_n : n \geq 0\}$, taking values $\prod_{j=1}^k \mathcal{S}_j$, is a Markov chain. What is its transition matrix?

7. Suppose that a rook and a bishop has been placed at opposite corners of an otherwise vacant chessboard. The pieces move simultaneously and independently of each other to squares where they can legally move. Note that we are allowing the pieces to come to the same square at the same time. Formulate this as a Markov chain. What is its transition matrix?

8. Let $\{\xi_n : n \geq 0\}$ be a sequence of independent identically distributed random variables taking values in $\{-1, 1\}$ with $\mathbb{P}(\xi_n = 1) = p = 1 - \mathbb{P}(\xi_n = -1)$. Let $X_n = \xi_n \xi_{n+1}$, for all $n \geq 0$. Is $\{X_n : n \geq 0\}$ a Markov chain?

9. Let $\{\xi_n : n \geq 1\}$ be a sequence of independent identically distributed integer-valued random variables. Let $S_0 = 0$ and for $n \geq 1$, $S_n = \sum_{i=1}^n \xi_i$. Also define $X_0 = 0$, $X_1 = \xi_1$ and for $n \geq 2$, $X_n = \xi_n + \xi_{n-1}$ and $Y_0 = 0$ and for $n \geq 1$, $Y_n = \sum_{k=1}^n S_k$. Which of the following constitute Markov chains : (a) $\{X_n : n \geq 0\}$, (b) $\{Y_n : n \geq 0\}$ and (c) the sequence of pairs $\{(S_n, Y_n) : n \geq 0\}$?

10. Find an example of a Markov chain $\{X_n : n \geq 0\}$ on the state space $\mathcal{S}$ and $x_0, x_2 \in \mathcal{S}$ and $A_1 \subseteq \mathcal{S}$ (having at least two elements) such that

$$\mathbb{P}(X_2 = x_2 \mid X_1 \in A_1, X_0 = x_0) \neq \mathbb{P}(X_2 = x_2 \mid X_1 \in A_1).$$

11. Consider the chain defined in Exercise 5. Set $A_0 = \{HHT, HTT, THT, TTT\}$, $A_1 = \{HTH, TTH\}$, $A_2 = \{THH\}$ and $A_3 = \{HHH\}$. We define a new process, called *aggregated process*, $\{Y_n : n \geq 0\}$ as follows: for any $i \in \{0, 1, 2, 3\}$, define $Y_n = i$ if $X_n \in A_i$. Show that the aggregated process is a Markov chain on $\{0, 1, 2, 3\}$. Find its transition matrix.

12. Show that not all stochastic matrices can be achieved as $\mathbf{P}^2$.

13. Set $Z_0 = 0$ and $Z_n = \max\{X_1, X_2, \dots, X_n\}$ for $n \geq 1$ where X_i are **iid** random variables taking values in $\{0, 1, 2, \dots\}$. Show that $\{Z_n : n \geq 1\}$ is a Markov chain. Find $p_{ij}^{(n)}$.

14. Consider a branching process $\{Z_n\}$. Suppose the offspring distribution is $\mathbb{P}(\xi = 0) = 0.2, \mathbb{P}(\xi = 1) = 0.5, \mathbb{P}(\xi = 2) = 0.3$.

1. Compute the transition probabilities p_{0j}, p_{1j}, p_{2j} for $j = 0, 1, 2, 3, 4$.
2. If the population starts with a single individual ($Z_0 = 1$), what is the probability that the population is extinct ($Z_1 = 0$) after one generation?
3. What is the probability that the population has exactly 2 individuals after one generation?

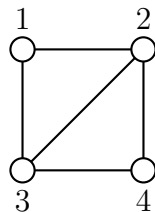
15. Consider the chain defined in Exercise 5. Compute $p_{ij}^{(n)}$. Consider the aggregated chain defined in Exercise 11. Can you find $p_{ij}^{(n)}$?

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Exercise Set 5.

1. Consider a random walk on the following graph:



- a) Write the transition matrix.
- b) If the walker starts at vertex 1, i.e., $\mathbb{P}(X_0 = 1) = 1$, what is the probability distribution after 2 steps, i.e., find $\mathbb{P}(X_2 = i)$ for $i = 1, 2, 3, 4$?
- c) Is this chain doubly stochastic? Explain.
2. For any Markov chain $\{X_n : n \geq 0\}$ with transition matrix $\mathbf{P}$, show that $\{Y_n := X_{nk} : n \geq 0\}$ is also a Markov chain for any $k \geq 2$. Find its transition matrix. Show by an example that the chain $\{Y_n : n \geq 0\}$ may not be irreducible even if the original chain is irreducible.
3. Consider a Markov chain on the state space $\mathcal{S} = \{1, 2, 3, 4\}$ with transition matrix

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{3}{4} & 0 & 0 & \frac{1}{4} \\ \frac{2}{3} & 0 & 0 & \frac{1}{3} \\ 0 & \frac{1}{2} & \frac{2}{3} & 0 \end{bmatrix} \end{matrix}.$$

- a) Find a recurrence relation between $p_{11}^{(n)}$ and hence find $p_{11}^{(n)}$.
- b) Compute $f_{11}^{(n)}$. Show that the state is positive recurrent. Compute $\mu_1 = \sum_{n=0}^{\infty} n f_{11}^{(n)}$.
4. Consider the random walk on a graph $G = (V, E)$. Let $i, j \in V$.

a) For $i_1, i_2, \dots, i_{n-1} \in V$, show that

$$\begin{aligned} & \deg(i) \mathbb{P}(X_1 = i_1, X_2 = i_2, \dots, X_{n-1} = i_{n-1}, X_n = j \mid X_0 = i) \\ &= \deg(j) \mathbb{P}(X_1 = i_{n-1}, X_{n-2} = i_2, \dots, X_{n-1} = i_1, X_n = i \mid X_0 = j) \end{aligned}$$

where $\deg(u)$ is the degree of the vertex u .

b) For any $n \geq 1$, show that

$$\deg(i) p_{ij}^{(n)} = \deg(j) p_{ji}^{(n)}.$$

5. For any $i, j \in \mathcal{S}$, show that

$$\sup_{n \geq 1} p_{ij}^{(n)} \leq f_{ij}^* \leq \sum_{n=1}^{\infty} p_{ij}^{(n)}.$$

6. Suppose that $i \rightsquigarrow j$. Define the event E_{ij} that chain visits j before returning to i starting from i . More precisely, for $n \geq 1$, set

$$g_{ij}^{(n)} = \mathbb{P}(X_n = j, X_t \neq i \text{ for } 1 \leq t < n \mid X_0 = i).$$

Show that $\mathbb{P}(E_{ij} \mid X_0 = i) = \sum_{n=1}^{\infty} g_{ij}^{(n)} > 0$.

7. Suppose $\{\epsilon_i : i \geq 1\}$ be **iid** random variable taking values in $\mathbb{Z}$ such that $\mathbb{E}(\epsilon_1) \neq 0$. Let $S_0 = 0$ and $S_n = \sum_{i=1}^n \epsilon_i$ for $n \geq 1$. Show that the chain $\{S_n : n \geq 0\}$ must be transient.

[Hint: Strong law of large numbers.]

8. For states i, j and any $n, k_0 \geq 1$, show that

$$\begin{aligned} & \mathbb{P}(X_t \neq j \text{ for } 1 \leq t \leq n + k_0 \mid X_0 = i) \\ & \leq \mathbb{P}(X_t \neq j \text{ for } 1 \leq t \leq n \mid X_0 = i) \sup\{\mathbb{P}(X_t \neq j \text{ for } 1 \leq t \leq k_0 \mid X_0 = k) : k \neq j\}. \end{aligned}$$

9. For states $i, j \in \mathcal{S}$ with $i \rightsquigarrow j$ in a Markov chain having N states show that there exists $n \leq N - 1$ such that $p_{ij}^{(n)} > 0$.

10. For a positive recurrent state i , show that

$$\mathbb{E}(\tau_i \mid X_0 = j) < \infty$$

for any j such that $i \rightsquigarrow j$.

11. Consider an irreducible aperiodic Markov chain.

- a) For any pair of states $i, j \in \mathcal{S}$, show that there exists $M = M(i, j)$ such that $p_{ij}^{(n)} > 0$ for all $n \geq M$.
- b) If the chain has N states, show that there exists $M = M(N)$ such that $p_{ij}^{(n)} > 0$ for all states i, j and $n \geq M$.
- c) For Exercise 1, explicitly come up with a value M such that $p_{ij}^{(n)} > 0$ for any $i, j \in \mathcal{S}$ and $n \geq M$.

12. Suppose $\{X_n : n \geq 0\}$ is a Markov chain. For some integer $K \geq 1$ and a state i_0 , suppose that

$$\inf\{p_{i,i_0}^{(K)} : i \in \mathcal{S}\} > 0.$$

Show that,

$$\mathbb{P}(\tau_{i_0} > n \mid X_0 = i) \leq C_1 \exp(-C_2 n)$$

for some $C_1, C_2 > 0$ and for any $i \in \mathcal{S}$ where

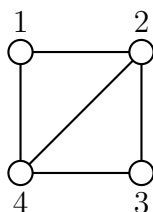
$$\tau_k = \inf\{n \geq 1 : X_n = k\}.$$

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Exercise Set 6.

1. Consider a random walk on the following graph:



Find the probabilities f_{ij}^* for $i, j \in \mathcal{S}$ and also $\mu_i = \mathbb{E}(\tau_i \mid X_0 = i)$.

2. Consider a Markov chain on the state space $\mathcal{S} = \{1, 2, 3, 4\}$ with transition matrix

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{3}{4} & 0 & 0 & \frac{1}{4} \\ \frac{2}{3} & 0 & 0 & \frac{1}{3} \\ 0 & \frac{1}{2} & \frac{2}{3} & 0 \end{bmatrix} \end{matrix}.$$

- a) Find the expected return times for each of the states 1, 2, 3, 4.
- b) What is the probability that the chain starting from 1, returns to 1 before visiting the state 4.
- b) What is the probability that the chain starting from 1, visits the state 3 at least twice before returning to state 1.
3. Consider two gamblers A and B, A with initial capital Rs. a and B with initial capital Rs. b . At each time point, first A tosses a coin with success probability p and then B tosses the same coin. If A gets success and B gets failure, then A wins Rs. 1 from B; if A gets failures and B gets success, then B wins Rs. 1 from A. In other cases, no exchange of money happens and the same procedure is repeated. What is the probability that A wins?

4. Consider the Markov Chain on $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$ with the transition matrix $\mathbf{P}$ given by

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} & \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{2}{5} & \frac{3}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{6} & \frac{1}{3} & 0 & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & 0 & \frac{1}{8} & 0 \end{bmatrix} \end{matrix}.$$

- a) Find the values of f_{33}^* and f_{44}^* .
- b) What is the probability that the chain starting from 3, visits the state 4 before visiting state 6.
- b) Let $\mathcal{C} = \{1, 2\}$. Show that $\mathcal{C}$ is a closed set. Define $\tau_{\mathcal{C}}$ as the first visiting time to $\mathcal{C}$, i.e.,

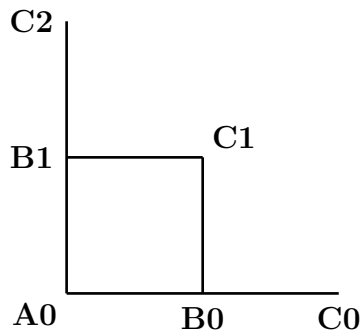
$$\tau_{\mathcal{C}} = \inf\{n \geq 0 : X_n \in \mathcal{C}\}.$$

Define $m_i = \mathbb{E}(\tau_{\mathcal{C}} \mid X_0 = i)$. For $i = 3, 4, 5, 6$, show that m_i satisfies the linear equations

$$m_i = 1 + \sum_{j=3}^6 p_{ij} m_j.$$

Solve the equations.

5. Consider a random walk on the graph



with the modifications that the vertices **C0**, **C1** and **C2** are absorbing states. Find the absorption probabilities at **C0** starting from any vertex.

6. Suppose a fair coin is being flipped until either of the patterns HTH or HHH appear. Find the probability that HTH appears before HHH.

7. You are playing a game which consists of tossing a coin until either one of the patterns of success and failures, one chosen by you and other by your opponent, appear. Assuming that the coin tosses are fair, show that for any pattern of length 3 chosen by your opponent, you can choose a pattern of length 3 so that the game is favourable to you.

[Hint : Representing H by 1 and T by 0, for a pattern $\epsilon_1\epsilon_2\epsilon_3$, chosen by your opponent, choose the pattern $(1 - \epsilon_2)\epsilon_1, \epsilon_2]$

8. Consider the Markov Chain on $\mathcal{S} = \{1, 2, 3, 4\}$ with the transition matrix $\mathbf{P}$ given by

$$\mathbf{P} = \begin{array}{c} \begin{array}{cccc} & 1 & 2 & 3 & 4 \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{bmatrix} \frac{1}{2} \\ \frac{2}{5} \\ 0 \\ \frac{1}{4} \end{bmatrix} & \begin{bmatrix} \frac{1}{2} \\ \frac{3}{5} \\ \frac{1}{3} \\ \frac{1}{4} \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ \frac{1}{3} \\ \frac{1}{4} \end{bmatrix} & \begin{bmatrix} 0 \\ 0 \\ \frac{1}{3} \\ \frac{1}{4} \end{bmatrix} \end{array} \end{array}.$$

Find all the stationary distributions.

9. Consider the Markov Chain on $\mathcal{S} = \{1, 2, 3, 4\}$ with the transition matrix $\mathbf{P}$ given by

$$\mathbf{P} = \begin{array}{c} \begin{array}{cccc} & 1 & 2 & 3 & 4 \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{bmatrix} \frac{1}{3} \\ \frac{3}{5} \\ 0 \\ \frac{1}{4} \end{bmatrix} & \begin{bmatrix} \frac{1}{3} \\ \frac{2}{5} \\ \frac{1}{3} \\ \frac{1}{4} \end{bmatrix} & \begin{bmatrix} 0 \\ \frac{1}{5} \\ \frac{1}{3} \\ \frac{1}{4} \end{bmatrix} & \begin{bmatrix} \frac{1}{3} \\ 0 \\ \frac{1}{3} \\ \frac{1}{4} \end{bmatrix} \end{array} \end{array}.$$

Find all the stationary distributions.

10. Consider two independent simple symmetric random walks X_n, Y_n . Consider the joint random walk restricted to $X_n \geq 0, Y_n \geq 0, X_n + Y_n \leq 10$. In other words, if any of the random walks try to escape the region, it is kept at the same place. Can you find the stationary distribution?

11. Consider an irreducible chain having finitely many states. Suppose that for a fixed state $i_0 \in \mathcal{S}$, we have $p_{ii_0} > 0$ for all $i \in \mathcal{S}$. Show that if

$$p_{i_0i}p_{ij}p_{ji_0} = p_{i_0j}p_{ji}p_{ii_0} \text{ for all } i, j \in \mathcal{S},$$

then there is a stationary distribution π and the chain is reversible in π .

[Hint: take $\pi(i) = Cp_{i_0i}/p_{ii_0}$ for a suitable choice of C]